

Randomised Projection

idea: Compress high dimension data to lower dimension to reduce computation/storage

Approach: map the original data $X \in \mathbb{R}^d$ into $\mathbb{R}^r$, with $r < d$, while maintain their pairwise distance, unchanged (or at least approximately)

projection map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^r$.

Let the Original Data lies in $\mathbb{R}^d \ni x$, and S_r an r -dimensional sub space of $\mathbb{R}^d$, with orthonormal basis $(v_1, v_2, \dots, v_r)$ spanning S_r ,

A linear projection map $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^r$ is defined as

$$y = \phi(x) = (\langle x, v_1 \rangle, \langle x, v_2 \rangle, \dots, \langle x, v_r \rangle)^T$$
$$= \begin{pmatrix} -v_1^T - \\ -v_2^T - \\ \vdots - \\ -v_r^T - \end{pmatrix} x = \Phi x, \quad \Phi \in \mathbb{R}^{r \times d}, \quad y \in \mathbb{R}^r.$$

random projection map ϕ , is when S_r is a random r -dim subspace.

Approx distance. preservation.

$y_i = \Phi x_i$, we wish for small $\epsilon > 0$

$$(1-\epsilon) \|x_i - x_j\|_2^2 \leq \|y_i - y_j\|_2^2 \leq (1+\epsilon) \|x_i - x_j\|_2^2, \quad \forall i, j \in \{1, \dots, n\}.$$

Gaussian Projectors

Consider random projection $\phi = \begin{pmatrix} -u_1^T \\ -u_2^T \\ \vdots \\ -u_r^T \end{pmatrix}$ where $u_i \in \mathbb{R}^d$ and $u_i \stackrel{\text{i.i.d.}}{\sim} N(0, 2\alpha)$ spherical Gaussian vectors.

Then for $v \in \mathbb{R}^d$, $\exists c > 0$ s.t. $\forall \epsilon \in (0, 1)$

$$P(|\|\phi v\|_2 - \sqrt{r}\|v\|_2| \geq \epsilon \sqrt{r}\|v\|_2) \leq 3e^{-c\epsilon^2}.$$

Remarks:

- The projected vector ϕv will be close to $\sqrt{r}\|v\|$ length with high probability

proof: $\phi_{ij} \sim N(0, 1) \Rightarrow \phi_{ij} v_j \sim N(0, v_j^2) \Rightarrow \sum_j \phi_{ij} v_j \sim N(0, \sum_j v_j^2)$

$$\text{thus } \frac{1}{\|v\|_2} \sum_j \phi_{ij} v_j = \frac{1}{\|v\|_2} (\phi v)_i \sim N(0, 1) \Rightarrow \frac{1}{\|v\|_2} \phi v \sim N(0, 2\alpha)$$

using Gaussian Annulus Theorem.

$$P\left(\left|\left\|\frac{1}{\|v\|_2} \phi v\right\|_2 - \sqrt{r}\right| \geq \epsilon \sqrt{r}\right) \leq 3e^{-c\epsilon^2}.$$

$$\Rightarrow P\left(\left|\|\phi v\|_2 - \sqrt{r}\|v\|_2\right| \geq \epsilon \sqrt{r}\|v\|_2\right) \leq 3e^{-c\epsilon^2}.$$

Johnson - Lindenstrauss Lemma,

let $x_1, \dots, x_n \in \mathbb{R}^d$, and Gaussian random projection $\phi = \begin{pmatrix} -u_1^T \\ -u_2^T \\ \vdots \\ -u_r^T \end{pmatrix}$

then $\exists c > 0$ s.t. $\forall \epsilon \in (0, 1)$ and $\delta > 0$, if

$r \geq \frac{c}{\epsilon^2} \log\left(\frac{5n^2}{\delta}\right)$, then

$$P\left(\left|\|\phi(x_j - x_k)\|_2 - \sqrt{r}\|x_j - x_k\|_2\right| \leq \epsilon \sqrt{r}\|x_j - x_k\|_2\right) \geq 1 - \delta, \forall j, k$$

Remarks:

the JL lemma states that the distance between projected point

$$(1 - \epsilon)\sqrt{r}\|x_j - x_k\|_2 \leq \|\phi(x_j - x_k)\|_2 \leq (1 + \epsilon)\sqrt{r}\|x_j - x_k\|_2.$$

with high probability.

proof:

let $V_{ij} = x_i - x_j$, from Gaussian projection theorem, we know $\exists C_{ij}, \epsilon_{ij} \in (0, 1)$ s.t.

$$P(|\|\phi V_{ij}\| - \sqrt{r} \|V\|_2| \geq \epsilon_{ij} \sqrt{r} \|V\|_2) \leq 3e^{-C_{ij} \epsilon_{ij}^2}$$

to show JL lemma, we need a union bound of event $\epsilon_{ij}: |\|\phi V_{ij}\| - \sqrt{r} \|V\|_2| \geq \epsilon_{ij} \sqrt{r} \|V\|_2$ so that above is true for $\forall i, j$.

$$P(\bigcup_{i,j} \epsilon_{ij}) \leq \sum_{i,j} P(\epsilon_{ij}) \leq \sum_{i,j} 3e^{-C_{ij} \epsilon_{ij}^2} \leq \binom{n}{2} e^{-C \epsilon^2}$$

where $C = \min\{C_{ij}\}$, $\epsilon = \min\{\epsilon_{ij}\}$, $\sum_{i,j}$ has $\binom{n}{2}$ terms.

$$\text{thus } P(\bigcup_{i,j} \epsilon_{ij}) \leq \binom{n}{2} e^{-C \epsilon^2} \leq 3n^2 e^{-C \epsilon^2} \leq \delta$$

$$\Rightarrow \delta e^{C \epsilon^2} \geq 3n^2 \Rightarrow r \geq \frac{1}{\epsilon^2 C} \cdot \log(3n^2/\delta)$$

Computing Random projection.

Direct computing require $O(nrd)$ time, n is sample size

idea: if $X = (x_1, \dots, x_n)^T$ has iid entry, we can randomly select r entry and approximate $\|X\|_2^2 \approx \frac{d}{r} \sum_{k=1}^r V_{ik}^2$

$$\text{so } \phi := \sqrt{\frac{d}{r}} P \text{ s.t. } \|\phi X\|_2^2 \approx \|X\|_2^2$$

issue: X is usually not iid entry, but tend to be sparse.

Fix: Consider. if x is sparse (s-function peaks), it's Fourier transform would be smooth (sin function).

we can find similar transformation s.t. the transformed X is flat.

Hadamard transform $U_m \in 2^m \times 2^m$.

U_m is defined recursively as $U_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, $U_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} U_1 & U_1 \\ U_1 & -U_1 \end{pmatrix}$.

... $U_m = \frac{1}{\sqrt{2}} \begin{pmatrix} U_{m-1} & U_{m-1} \\ U_{m-1} & -U_{m-1} \end{pmatrix}$, each entry $\frac{1}{\sqrt{2^m}}$.

Remarks: • U_m is orthogonal., $U_m U_m^T = U_m^T U_m = I$.

• $U_m x$ can be computed using similar algorithm as FFT with $O(2^m \log 2^m)$ time.

• U_m maps sparse vector. to a flat vector.

• flatness can be further improved by randomization.

$D_{ij} = \begin{cases} \text{sign}(x_j) & P = 1/2 \\ -\text{sign}(x_j) & P = -1/2 \end{cases} \rightarrow \text{independ } U_m D \rightarrow \text{randomized Hadamard transform.}$
independent random sign flip

Lemma - $U_m D x$ is flat.

$$\forall x \in \mathbb{R}^d, P(\|U D x\|_\infty \geq t) \leq 2d e^{-\frac{dt^2}{2\|x\|_2^2}}.$$

proof: $u_i := (U D x)_i = \sum_{j=1}^d a_j x_j$, $a_j = \begin{cases} \frac{1}{\sqrt{d}}, & P = 1/2 \\ -\frac{1}{\sqrt{d}}, & P = -1/2. \end{cases}$

for each entry i

$$P(\|u\| \geq t) \leq 2 \exp\left(-\frac{t^2}{\sum_{j=1}^d (x_j \frac{1}{\sqrt{d}})^2}\right) \quad (\text{Hoeffding}).$$

$$= 2 \exp\left(-\frac{dt^2}{2\|x\|_2^2}\right).$$

$$P(\|u\|_\infty \geq t) \leq d P(\|u\| \geq t) \leq 2d \exp\left(-\frac{dt^2}{2\|x\|_2^2}\right)$$

Remark. $\frac{1}{\sqrt{d}} \|U D x\|_2 = \frac{1}{\sqrt{d}} \|x\|$, if $t = k \frac{\|x\|_2}{\sqrt{d}}$, then the probability of $\|U D x\|_2$ not flat (one component is very large compared to other $\leftrightarrow \|u\|_\infty = k \frac{\|x\|_2}{\sqrt{d}}$ with large k) would be small.

Fast JLT

Since UDx is flat with high probability, then $\phi = \sqrt{\frac{q}{r}} PHD$ would preserve the norm with high prob.

$$\text{as } \|\phi x\|_2 = \sqrt{\frac{q}{r}} \|PHDx\|_2 \approx \|UDx\|_2 = \|x\|_2$$

Remark: the computation cost for FJLT is

$$O\left(\underbrace{nd \ln d}_{Ux} + \underbrace{\frac{d}{r}}_{P \text{ and } D}\right) = O(nd \ln d)$$